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Impacts of general agents' customer relationship-building behaviors on trust, satisfaction, and renewal intention: A PLS-SEM and NCA approach

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ABSTRACT

This study investigates how insurance general agents' relationship-building behaviors influence customer trust, satisfaction, and renewal intention. Grounded in relationship marketing and social exchange theories, five key behaviors—approach, inducement, common grounding, networking, and communication—were tested using Partial Least Squares Structural Equation Modeling (PLS-SEM) and Necessary Condition Analysis (NCA). Data was collected from 253 insurance customers. The findings revealed that approach, common grounding, and communication significantly enhance trust. Only communication positively affects satisfaction. Furthermore, trust exerts a positive influence on both satisfaction and renewal intention, while satisfaction also emerges as a strong driver of renewal decisions. Mediation analyses indicate that trust partially mediates the relationship between communication and satisfaction, underscoring the importance of consistent and transparent information exchange in building customer trust. Moreover, satisfaction partially mediates the link between trust and renewal intention, reinforcing the pivotal role of emotional attachment for long-term contract retention. The NCA results show communication, trust, and satisfaction to be "necessary" factors, implying that without reaching certain thresholds in these variables, customers are unlikely to renew insurance contracts. These findings highlight the essential nature of proactive, empathetic, and clear communication strategies in fostering enduring customer loyalty. From a managerial perspective, insurers should prioritize clear, timely, and empathetic communication and foster trust-based engagement to optimize service delivery.

1. Introduction

As digital technology continues to advance and the contact points between companies and customers increase, firms must adopt new strategies to build relationships with their customers (Al Masud et al., 2024; Vander Schee et al., 2020). As a result, online subscription channels are expanding; however, face-to-face sales channels remain the preferred method for many customers (Singh & Shah, 2024). The insurance industry exemplifies this trend. Given the intangible and complex nature of insurance products, customers often seek clearer information and a sense of psychological security through in-person interactions (Agyei et al., 2020). The significance of these face-to-face channels is further underscored by the fact that trust (hereafter "trust") and engagement between customers and insurance agents play a crucial role in purchasing decisions and fostering renewal intention (Singh & Shah, 2024).

Insurance agents are more than just product sellers; they play a vital role in establishing long-term relationships with customers, fostering trust, and enhancing satisfaction (Khan et al., 2022). Their relationship-building efforts contribute to customers' psychological stability and create positive purchasing experiences, ultimately leading to higher satisfaction and increased likelihood of contract renewal (Gao et al., 2022; Gwinner et al., 1998; Khan et al., 2022). The insurance market has become increasingly competitive due to the expansion of General Agency (GA) channels and the weakening of tied-agent channels. Excessive commission competition and frequent agent turnover have led to a decline in policy retention rates (Korea Insurance Research Institute [KIRI], 2025). The ongoing focus on short-term sales performance by insurers has resulted in rising policy switching rates and decreased contract stability, negatively impacting Contractual Service Margin (CSM) (KIRI, 2025; Tomiris & Cho, 2024). CSM serves as a key indicator of an insurer's future profitability, and declining retention rates directly

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affect the long-term financial sustainability of insurance firms (Rather & Hollebeek, 2019). Therefore, the insurance industry must adopt practical countermeasures, including long-term customer retention strategies and strengthening agents' relationship-building efforts to ensure sustainable growth (Kim et al., 2021). However, despite the seminal theoretical foundation laid by Morgan and Hunt (1994), recent literature reviews and empirical studies (Rather & Hollebeek, 2019; Makwata & Mwanzia, 2024; Gao et al., 2022) continue to highlight a paucity of systematic, context-specific analyses on how insurance agents' relationship-building behaviors influence trust, satisfaction, and renewal intention.

Previous studies have highlighted the significance of insurance' relationship-building behaviors through the lens of relationship marketing theory and social exchange theory (SET). Customer relationship-building behaviors play a crucial role in influencing renewal intention, both directly and indirectly through trust and satisfaction. For instance, recent research indicates that relationship quality significantly impacts repurchase intention via satisfaction (Chiu et al., 2021). Insurance agents' relationship-building behaviors can be understood through the lens of relationship marketing theory (Berry, 1995; Morgan & Hunt, 1994) and social exchange theory (Blau, 2017), as demonstrated in recent studies on e-brand experience and relationship quality (Chen & Qasim, 2021). Despite extensive research on customer relationship-building behaviors across various service contexts, limited empirical studies have specifically examined the impact of customer relationship management strategies on loyalty in the insurance sector. Recent findings highlight that customer interaction and target marketing significantly enhance customer loyalty, while customer service delivery may have a weaker influence (Makwata & Mwanzia, 2024). Furthermore, in-person interactions remain vital for fostering trust and emotional connections, even in the digital era, as face-to-face communication provides richer social cues and deeper engagement compared to digital alternatives (Gruber et al., 2022). First, there are limited empirical studies examining the impact of insurance Agents' relationship-building behaviors on trust and renewal intention (Meira & Hancer, 2021; Rather & Hollebeek, 2019). Second, little research has been conducted on the crucial role of key relationship-building factors in the process of fostering trust, satisfaction, and customers' renewal intention (Castillo-Alarcón & Valenzuela-Fernández, 2024). Third, few studies have empirically demonstrated that face-to-face interactions remain essential even in the digital era (Dzogbenuku et al., 2022).

To bridge these gaps, this study makes the following three key contributions. First, we examine the impact of insurance agents' relationship-building behaviors on trust, satisfaction, and renewal intentions using relationship marketing and social exchange theory. To explore the underlying mechanisms, we incorporate five auxiliary theories. Trust-building theory illustrates how consistency and transparency act as signals that minimize perceived risk and instill confidence. Motivation theory connects inducement behaviors to intrinsic and extrinsic incentives, which drive customer engagement. Empathy theory emphasizes how shared understanding fosters emotional resonance, deepening relational bonds. Social network theory positions networking as a means of leveraging social connections to broaden access to information and resources. Lastly, communication accommodation theory highlights how adaptive verbal and nonverbal communication enhances clarity and strengthens perceived competence (Alam et al., 2021; Gao et al., 2022; Holmström et al., 2024; Li et al., 2023; Templeton et al., 2023). Second, we adopt a complementary methodological approach that combines Partial Least Squares Structural Equation Modeling (PLS-SEM) and Necessary Condition Analysis (NCA), as demonstrated by Sukhov et al. (2022), to identify both sufficient and necessary conditions for customer loyalty. This integrated strategy allows for a deeper exploration of factors that are not only important but essential for achieving high renewal intention in insurance settings. Third, by comparing the results derived from PLS-SEM and NCA, we contribute to the advancement of theory and practice by distinguishing between 'should-have' and 'must-have' behaviors in agent-customer relationship management (Sukhov et al., 2022). Building on previous research, this study examines the structural relationship between insurance agents' relationship-building behaviors and trust and renewal intention. Furthermore, using Necessary Condition Analysis (NCA), it empirically demonstrates that specific relationship-building behaviors are essential conditions for fostering customer loyalty. Specifically, this study addresses the research question: how do insurance agents' relationship-building behaviors influence trust, satisfaction, and renewal intention?

This study makes a distinct contribution to the literature by not only identifying statistically significant factors through PLS-SEM but also uncovering behaviorally indispensable conditions using Necessary Condition Analysis (NCA). By doing so, it advances prior research on relationship marketing and Social Exchange Theory (SET). Unlike previous studies that primarily examined either relational behaviors or their sufficiency, our research integrates necessity logic to illustrate that behaviors such as communication are not merely advantageous but fundamentally essential for fostering customer loyalty within the insurance sector.

To enhance clarity in understanding the research process, the Research Flowchart visually maps out the critical stages of this study

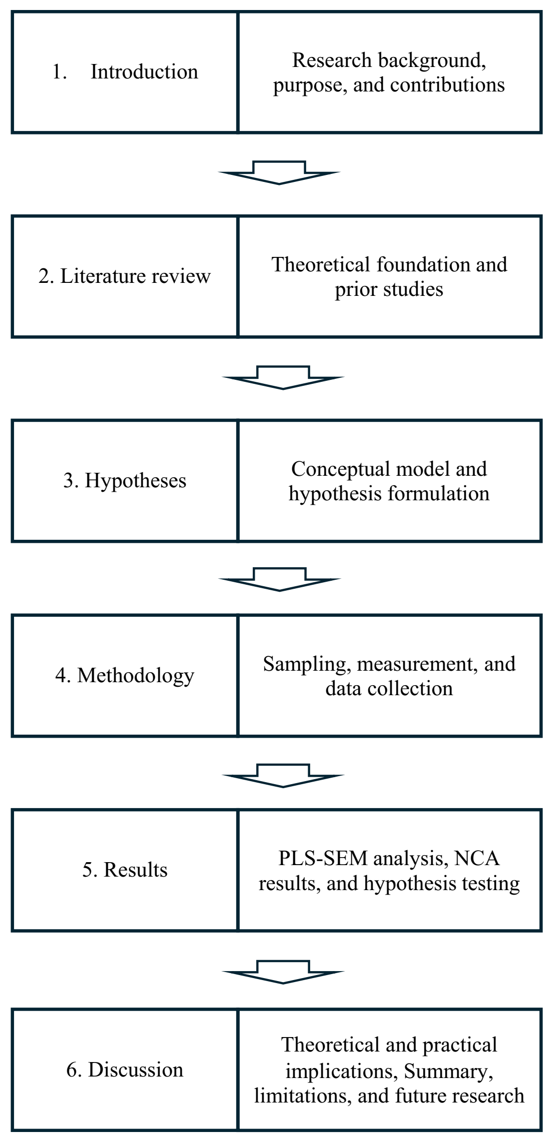

Fig. 1. Research flowchart.

(see Fig. 1). It systematically presents key steps, including theoretical foundation, hypothesis formulation, survey construction, data gathering, statistical analysis, and the interpretation of findings, ensuring a structured overview of the study's methodological approach.

2. Literature review

2.1. Relationship marketing theory and social exchange theory

The customer relationship-building behavior of insurance agents can be understood through the lens of relationship marketing theory and SET (Arditto et al., 2020; Gremler et al., 2020). At its core, Morgan and Hunt's (1994) Trust-Commitment theory posits that trust and commitment are the primary drivers of enduring exchanges: trust reduces perceived risk and uncertainty, while commitment reflects a long-term orientation toward the relationship. Relationship marketing theory emphasizes that businesses should focus on fostering long-term relationships with customers rather than merely engaging in short-term transactions (Berry, 1995; Morgan & Hunt, 1994). This implies that insurance agents should actively work to build trust, enhance satisfaction, and cultivate loyalty through ongoing interactions (Gwinner et al., 1998; Palmatier et al., 2006).

According to relationship marketing theory, customers do not merely purchase products or services; rather, they evaluate the overall benefits gained from their relationship with a company when deciding whether to continue their transactions (Palmatier et al., 2006). In this context, relationship-building behaviors—such as proactive engagement, effective communication, inducements, empathy, and networking—play a crucial role in fostering trust, enhancing satisfaction, and ultimately influencing re-contracting intentions (Berry, 1995; Gwinner et al., 1998). For instance, when insurance agents maintain clear and consistent communication, provide personalized information, and actively manage relationships, trust and satisfaction are more likely to increase.

SET posits that human relationships are formed and sustained through mutual reciprocity and expected rewards (Blau, 2017; Kilroy et al., 2023). This perspective suggests that the relationship between an insurance agent and a customer extends beyond a simple contractual agreement—it is an ongoing exchange in which both parties anticipate mutual benefits (Morgan & Hunt, 1994). Customers assess the value of maintaining the relationship based on the trust, information, and emotional support provided by the insurance agent, while insurance agents seek to enhance their sales performance by fostering customer satisfaction and renewal intention (Palmatier et al., 2006). Notably, SET explains that when customers positively perceive an insurance agent's relationship-building behaviors, they develop a strong commitment to maintaining the relationship based on trust and satisfaction (Blau, 2017; Gwinner et al., 1998). Therefore, if customers trust an insurance agent and experience both emotional bonds and tangible benefits through continuous interactions, they are more likely to renew their insurance policy.

2.2. Customer relationship building behavior

Customer relationship building behavior is a strategic practice among insurance agents to establish and sustain long-term relationships with clients, significantly influencing trust, satisfaction, and renewal intentions (Morgan & Hunt, 1994; Palmatier et al., 2006). Strengthening trust through ongoing customer interaction and maintaining enduring relationships are crucial (Berry, 1995; Gwinner et al., 1998). This study examines customer relationship building behavior by categorizing it into five elements: approach, inducement, common grounding, networking, and communication (Lim et al., 2008; Schwepker Jr & Good, 2022).

Approach in insurance planning involves actively reaching out to customers to establish relationships (Verhoef & Lemon, 2013). This is crucial for building initial trust and serves as the first step for customers to rely on insurance agents (Nayak et al., 2024). A proactive approach not only engages customers but also sets the foundation for a lasting relationship and facilitates continuous interaction (Palmatier et al., 2006).

Inducement is the process of guiding customers to become interested in insurance products, and it plays a role in strengthening customers' purchase intentions (Hunt, 1997; Shahroodi et al., 2024). It is a process that emphasizes the benefits expected by customers beyond simple product descriptions, and includes strategies such as customized suggestions, additional service provision, and discounts (Gwinner et al., 1998). Effective inducement increases customer satisfaction and increases the possibility of maintaining long-term relationships (Berry, 1995).

Common grounding is the process of understanding customers' needs and forming emotional bonds, and it contributes to improving trust and satisfaction (Gao et al., 2022; Möller & Halinen, 2000). It is key to identifying problems from the customer's perspective and providing personalized services, and this is an important factor in customers' trust in their relationship with insurance agents. The stronger empathy, the more likely it is that customers will perceive insurance agents as trustworthy consultants, not just sellers (Gerlach et al., 2016; Nayak et al., 2024).

Networking refers to the behavior of insurance agents expanding their relationships with customers and maintaining continuous connections with them (Morgan & Hunt, 1994). It goes beyond simple individual customer management and includes the process of forming new relationships by utilizing customers' social networks (Palmatier et al., 2006). The more active the networking with customers, the more likely it is that the relationship will be sustainable, which leads to long-term loyalty of customers (Lin et al., 2025; Trzcielinski et al., 2024).

Communication refers to clear and consistent information exchange with customers and is a key element in building trust (Alam et al., 2021; Sharma & Patterson, 1999). Considering the complexity of insurance products, effective communication by designers plays an important role in increasing customer understanding and building trust (Ball et al., 2004; Leung & Chan, 2004). In addition, smooth communication reduces customer uncertainty and enhances the stability of relationships, thereby strengthening the renewal intention (Herjanto & Amin, 2020; Ruefenacht, 2018).

2.3. Trust

Trust in an exchange relationship means the belief that the words or promises of the other party are trustworthy and that they will faithfully fulfill their obligations (Starke, 2025; Wang et al., 2020). Trust describes an individual's rational evaluation and reflects a belief in reliability, dependability, and competence. Trust is defined as the degree of confidence or willingness to rely on the trust of the other party (Akrout et al., 2016). When assessing trust in an insurance agent, consider the history related to the work and closely examine how the insurance agent handled specific situations (Rajavi et al., 2019). An insurance agent who is perceived as fair is more likely to be trusted. This trust is essential for building a long-term and successful partnership between the customer and the insurance agent (Bhattacharyya et al., 2024).

2.4. Satisfaction

Satisfaction refers to the positive emotions and overall evaluation that customers feel after experiencing the service and interaction with the insurance agent (Oliver, 1999). It is determined by the difference between the service level expected by the customer and the actual service level provided, and the satisfaction increases as the customer experiences higher service quality than expected (Fornell et al., 1996). In the insurance industry, reliability, professionalism, communication style, and relationship-building behavior of insurance agents are important factors in forming customer satisfaction (Agyei et al., 2020).

High satisfaction strengthens the customer's renewal intention and contributes to attracting new customers by inducing a positive word-of-mouth effect (Rather & Hollebeek, 2019; Son et al., 2023).

3. Hypotheses

3.1. Customer relationship-building behavior and trust

Trust is a critical antecedent of long-term customer retention in high-involvement services such as insurance, where products are intangible and outcomes uncertain (Morgan & Hunt, 1994). Trust reflects a customer's willingness to rely on an agent's integrity, competence, and benevolence (Chaudhuri & Holbrook, 2001). Empirical studies show that specific relationship-building behaviors by agents serve as trust-enhancing signals. Proactive approach behaviors—such as timely outreach and strict adherence to appointments—convey reliability and respect for customers' time, thereby reducing perceived risk (Alam et al., 2021). Inducement behaviors, including personalized gifts and special offers, demonstrate commitment beyond mere transactions and bolster perceived sincerity (Shahroodi et al., 2024). Through common grounding—sharing personal anecdotes—agents build emotional rapport and reduce social distance, deepening trust (Li et al., 2024). Networking behaviors, such as connecting clients with peers or experts, showcase social capital and resourcefulness, further enhancing credibility (Gao et al., 2022). Finally, clear communication—characterized by transparent explanations and timely updates—directly strengthens perceptions of competence and integrity (Gün & Söyük, 2025). Collectively, these behaviors have been shown to significantly enhance trust in insurance agents.

Therefore, we propose the following hypotheses.

H1. Customer relationship-building behavior will positively influence trust.

H1a. Approach behavior will positively influence trust.

H1b. Inducement will positively influence trust.

H1c. Common grounding will positively influence trust.

H1d. Networking behavior will positively influence trust.

H1e. Communication behavior will positively influence trust.

3.2. Customer relationship-building behavior and satisfaction

Satisfaction is defined as a customer's overall affective response when service performance meets or exceeds expectations (Oliver, 1999). Satisfaction not only reflects positive evaluations but also drives loyalty behaviors such as repeat purchase and contract renewal (Hennig-Thurau et al., 2002). In insurance services—where complexity can lead to confusion—agents' relationship-building behaviors enhance perceived service quality by aligning interactions with customer needs (Yu, 2024). Approach behaviors create a welcoming environment and set positive expectations, leading to enhanced satisfaction (Alam et al., 2021). Inducement behaviors provide pleasant surprises that improve customers' emotional responses and satisfaction (Shahroodi et al., 2024). Common grounding personalizes interactions, making customers feel understood and valued, thereby heightening satisfaction (Li et al., 2024). Networking behaviors expand informational support, enabling customers to feel more informed and satisfied (Gao et al., 2022). Communication clarity reduces ambiguity and reinforces perceptions of professionalism, directly boosting satisfaction (Gün & Söyük, 2025). When these behaviors consistently exceed expectations, they lead to higher satisfaction, which in turn fosters loyalty and renewal intention (Hennig-Thurau et al., 2002).

Therefore, we propose the following hypotheses.

H2. Customer relationship-building behavior will positively influence customer satisfaction.

H2a. Approach behavior will positively influence customer satisfaction.

H2b. Inducement will positively influence customer satisfaction.

H2c. Common grounding will positively influence customer satisfaction.

H2d. Networking behavior will positively influence customer satisfaction.

H2e. Communication behavior will positively influence customer satisfaction.

3.3. Trust and satisfaction

Trust is built on the reliability, honesty, and professionalism of the service provider and serves as a key precursor to customer satisfaction (Morgan & Hunt, 1994; Oliver, 1999). When trust is established, customers experience reduced uncertainty about the service provider and accumulate positive interactions, leading to higher satisfaction (Chaudhuri & Holbrook, 2001; Rather & Hollebeek, 2019). Trust influences customer satisfaction through both service quality and relationship quality (Gün & Söyük, 2025; Hennig-Thurau et al., 2002). Particularly in high-involvement industries like insurance, customers tend to seek long-term relationships based on trust, further reinforcing their overall satisfaction (Bifkovics et al., 2024; Tran et al., 2024).

Based on previous studies, we hypothesize:

H3. Trust positively influences satisfaction.

3.4. Trust and renewal intention

Trust is a crucial factor in establishing a lasting relationship with a service provider and has a direct impact on contract renewal intentions (Liu et al., 2025; Tong et al., 2024). Renewal intention is a key factor in customer loyalty, reflecting their willingness to renew or extend contracts with the same service provider, especially in the insurance industry (Nayak et al., 2024; Singh & Shah, 2024). Customers with a high level of trust perceive the quality of service and honesty of the provider more positively, making them more likely to renew or maintain their contracts (Chaudhuri & Holbrook, 2001; Oliver, 1999; Tran & Vu, 2019). Additionally, trust serves as a key factor in reducing perceived risk and increasing customers' willingness to continue using the service (Hennig-Thurau et al., 2002; Rather & Hollebeek, 2019). Given the high-involvement nature of insurance products, building trust is essential. Enhancing trust through transparent information, prompt claims processing, and personalized services strengthens contract renewal intentions (Wu et al., 2022). For instance, tailored advice from insurance agents and efficient claims handling by insurance companies foster trust, ultimately increasing the likelihood of contract renewal (Skaf et al., 2024).

Therefore, we propose the following hypothesis.

H4. Trust positively influences intent to renewal intention.

3.5. Satisfaction and renewal intention

Customer satisfaction is influenced by the perception of service provider performance meeting or surpassing expectations, significantly impacting long-term relationship maintenance and renewal intention (Hennig-Thurau et al., 2002; Oliver, 1999). Satisfied customers are more inclined to renew contracts, trusting the provider and building positive experiences (Chaudhuri & Holbrook, 2001; Rather & Hollebeek, 2019; Tran & Vu, 2019). Previous studies have highlighted the significant impact of service quality and satisfaction on customer loyalty and renewal intention (Tran & Vu, 2019; Xiang et al., 2024). Specifically, in long-term contractual relationships like insurance services, higher customer satisfaction increases the likelihood of maintaining continuous contracts, fostering customer loyalty, and generating positive word-of-mouth effects (Dangaiso et al., 2024).

Thus, we propose the following hypothesis.

H5. Satisfaction positively influences renewal intention.

3.6. Mediation effects of satisfaction and trust

Drawing on Signaling Theory (Spence, 1978) and Expectation–Confirmation Theory (Oliver, 1980), this study posits two key psychological mediators—trust and satisfaction—in the pathway from relationship-building behaviors to customer renewal intention. In service contexts with high information asymmetry, agents' relational behaviors—such as transparent communication and empathy—signal reliability and reduce perceived risk. Gao et al. (2022) demonstrated that trust fully mediates the effect of service recovery strategies on customer satisfaction in the financial services industry, emphasizing its central role in translating interactional cues into satisfaction. According to Expectation–Confirmation Theory, when service performance meets or exceeds expectations, satisfaction increases, fostering positive behavioral intentions. Albarq (2023) showed that customer satisfaction significantly mediates the link between trust and loyalty in the banking sector, while trust itself also acts as a partial mediator between knowledge management and satisfaction.

Moreover, emerging research has shown that relationship-building behaviors, including communication, inducement, empathy, networking, and information sharing, can also exert direct effects on renewal intention. Gao et al. (2022) empirically demonstrated that communication and empathy significantly and directly influence customer loyalty in the insurance industry. Similarly, Meira and Hancer (2021) confirmed that relationship-building activities, encompassing inducement and networking behaviors, have a direct positive impact on repurchase intentions in service contexts. These findings suggest that both direct effects and mediated effects through trust and satisfaction must be simultaneously considered to fully capture the dynamics of customer renewal intention.

Thus, we propose the following hypothesis.

H6. Trust will mediate the relationship between relationship-building behaviors and satisfaction.

H7. Satisfaction will mediate the relationship between relationship-building behaviors and renewal intention.

Based on the hypotheses, the proposed model is shown in Fig. 2.

4. Methodology

4.1. Sampling and data collection

This study followed a positivist research philosophy and utilized a quantitative, cross-sectional survey design. Data was gathered from a nationwide panel of insurance customers aged 20 and older. Since the survey did not involve the collection of sensitive information, it was exempt from Institutional Review Board (IRB) review.

The participants were insurance customers who had purchased policies through agents and maintained their contracts. They were recruited from a nationwide online panel encompassing multiple insurance providers, ensuring a diverse and representative sample. This study utilized quota sampling to accurately represent the demographic composition of insurance consumers in Korea. The sample distribution was determined according to gender, age, and income level, based on official statistics from the Korea Insurance Research Institute (2024) and the Financial Supervisory Service (2024). For instance, female respondents comprised 58.1 % of the sample, closely aligning with the national average of approximately 55 %. Additionally, the highest proportion of respondents were in their 40s, which reflects the actual age distribution of insurance subscribers. This approach strengthens the external validity of the study. Specifically, we collected the data through the help of a professional survey company after defining the proportions by respondent strata based on key demographic—age group, gender, income bracket, and region—based on the latest national statistics for insurance customers. Once each quota cell (e.g., males aged 40–49 with monthly income ₩4–6 million) was filled, recruitment for that cell ceased. This approach is a non-probability (quota) sampling technique, which—while not strictly random—ensures that our sample mirrors the known population distribution, thereby enhancing external validity when true probability sampling is impractical.

The survey was conducted over a one-month period, from January 6, 2024, to February 5, 2024, using a web-based platform with the help of

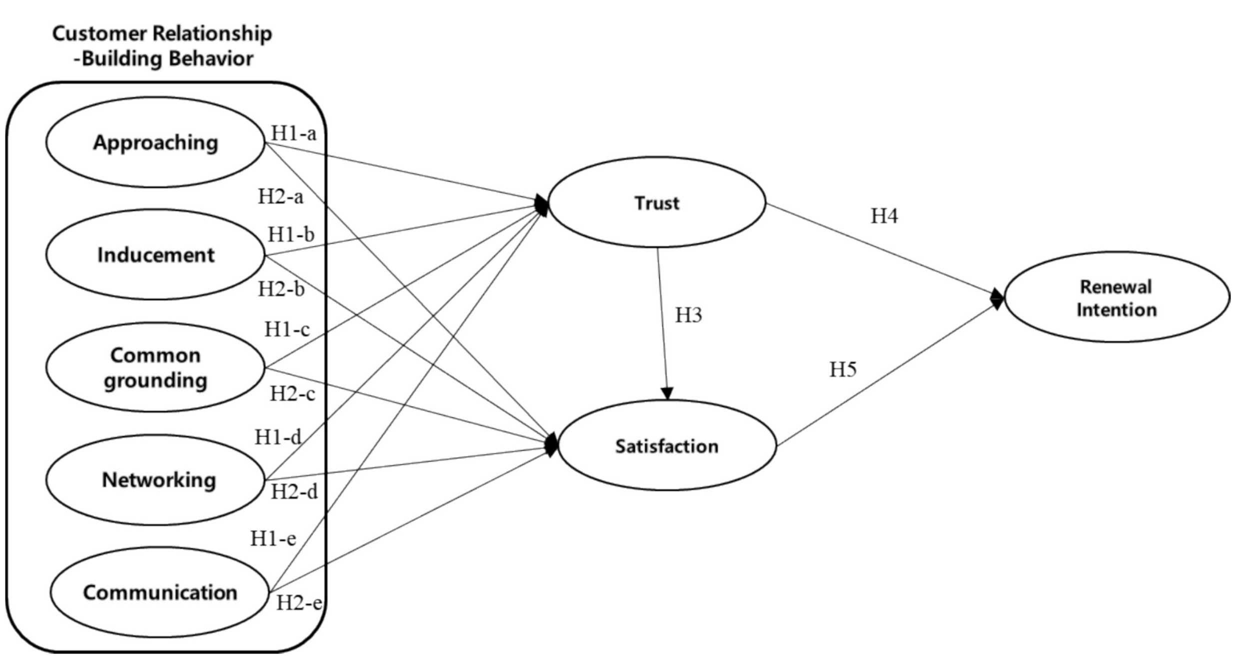

Fig. 2. Proposed model.

an online survey company. A total of 486 questionnaires were distributed to online panelists, with 253 completed responses used for analysis. We conducted a two-stage pre-test and employed quota-based sampling.

We conducted a two-stage pre-test: first with 20 internal respondents to refine wording, then a pilot with 50 panelists to assess completion time, item clarity, and technical issues. Feedback led to minor edits (e.g., reordering items, simplifying instructions). Next, the online survey company sent invitations via email, telephone, and KakaoTalk, including two reminders over a two-week period, and used a screening item to confirm at least one face-to-face interaction with an agent in the past year. To ensure data quality and optimize response rates, the online survey company monitored response times, excluding any submissions completed in less than one-third of the median response time. Furthermore, only responses from participants who completed the entire survey were included in the final analysis. As each demographic quota cell reached its capacity, it was automatically closed to maintain balanced representation across age, gender, income, and region. Out of 486 initial participants, 253 valid responses remained after applying quality control measures and fulfilling demographic quotas.

Following Memon et al. (2020)'s guidelines, the minimum required sample size was calculated as 103 (effect size = 0.15, α = 0.05, power = 0.8, number of predictors = 7) using the G*Power 3.1.9.7 program. Consequently, the sample size of 486 in this study is more than adequate. Additionally, the subgroup sample sizes—166 for the Junior College or below group and 360 for the University or above group—exceed the minimum threshold of 50, ensuring their suitability for multi-group analysis (Hair et al., 2017). Furthermore, based on Hair Jr et al. (2021)'s 10-times rule, which stipulates that the minimum sample size in a structural model should be at least ten times the number of paths leading to any construct, the current sample size is deemed sufficient.

Online surveys are widely utilized across various fields due to their cost-effectiveness, accessibility to large samples, well-structured sampling methods, and ease of follow-up (Kim et al., 2019). We administered the survey in Korean, reflecting the language of our target population. The original measurement items (adapted from English-language scales) underwent a rigorous forward–back translation process. A bilingual researcher translated the English items into Korean. Two Korean-English bilingual professors reviewed the Korean version for clarity and cultural relevance. A different bilingual expert translated the Korean draft back into English. We compared the back-translated English with the original to resolve discrepancies and ensure semantic equivalence. This process ensures that the Korean questionnaire faithfully captures the constructs of the original scales.

Additionally, we gathered information on respondents' insurance types and their frequency of interactions with agents. Life, health, and auto insurance comprised 42 %, 37 %, and 21 % of the total, respectively, with over 60 % reporting annual in-person meetings. These trends provide context for customer perceptions and reinforce the study's relational emphasis.

4.2. Measures

All constructs were measured using multi-items using 7-point Likert scales anchored by "1 = strongly disagree" and "7 = strongly agree". Customer relationship-building behaviors were measured as five sub-dimensions such as approach (5 items, Verhoef & Lemon, 2013), inducement (6 items, Chandon et al., 2000), common grounding (5 items, Hunt, 1997), networking (7 items, Möller & Halinen, 2000), and information communication (6 items, Sharma & Patterson, 1999). Trust was defined based on the professionalism and consulting reliability of insurance agents and measured using 6 items (Kim et al., 2019). Satisfaction refers to the overall satisfaction customers feel with the insurance agent's service measured using 3 items (Dawes, 2025). The renewal intention refers to the customer's renewal intention with the current insurance agent and measured using 2 items (Lafferty et al., 2002).

5. Results

5.1. Demographic profiles

The general characteristics of the 253 survey respondents are summarized in Table 1. The sample had a higher proportion of female respondents (58.1 %) compared to male respondents (41.9 %). The largest age group comprised individuals in their 40s (36.8 %), followed by those in their 50s (22.9 %) and 30s (20.6 %). In terms of marital status, most respondents were married (80.6 %). Regarding educational level, the highest proportion of respondents had obtained a four-year university degree (37.5 %), followed by those with a high school (24.5 %) and those who had completed junior college (19.8 %). In terms of income distribution, the largest group of respondents earned between 4 and 5.99 million won (34.4 %), followed by those earning >8 million won (20.6 %) and those earning between 3 and 3.99 million won (18.2 %). Occupationally, the most common category was self-employment (39.5 %), followed by sales and service (15.4 %) and public official (12.6 %). Finally, the types of insurance products held by respondents, non-life insurance (80.2 %), which include comprehensive insurance, indemnity insurance, auto insurance, and fire insurance, was more commonly held than life insurance (19.8 %), which includes whole life insurance, critical illness (CI) insurance, and variable life insurance.

The sample is predominantly female (58 %) and middle-aged (37 % in their 40s), demographics often linked to greater risk aversion and a preference for personalized advisory services. The significant proportion of self-employed respondents (39.5 %) indicates a reliance on professional guidance for navigating complex insurance decisions. Additionally, over 60 % reported at least annual face-to-face interactions with their agent, highlighting the importance of in-person relationship-building in fostering trust and customer retention.

Table 1
Demographic profiles.

Demographics	n	%
Gender		
Male	106	41.9
Female	147	58.1
Age		
20–29	12	4.7
30–39	52	20.6
40–49	93	36.8
50–59	58	22.9
over 60	38	15.0
Marital status		
Single	49	19.4
Married	204	80.6
Educational level		
Graduate high school	62	24.5
Junior college graduate	50	19.8
University	95	37.5
Graduate school	46	18.2
Monthly income (KRW: Million won)		
<2	8	3.2
2 ~ <3	28	11.1
3 ~ <4	46	18.2
4 ~ <6	87	34.4
6 ~ <8	32	12.6
>8	52	20.6
Job		
Administrative/office worker	1	0.4
Self-ownership	100	39.5
Sales/Service Jobs	39	15.4
Housewife	14	5.5
Technology/production job	3	1.2
Professional	12	4.7
Other	45	17.8
Public official	32	12.6
Insurance products signed up		
Life insurance	50	19.8
Non-life insurance	203	80.2

5.2. Measurement model analysis

The reliability and validity of all constructs were verified through measurement model analysis using SmartPLS 4.1 (see Table 2). The Cronbach's α values of all constructs were 0.943–0.984, and the composite reliability (CR) values were 0.963–0.992, which exceed the generally required standard (0.70 or higher), and the measured variables had high internal consistency (Hair et al., 2017). The factor loadings of all variables were 0.70 or higher, and the average variance extracted (AVE) values were 0.851–0.985, which met the criterion (0.50 or higher), establishing convergent validity (see Table 3). As seen in the Fornell-Larcker criterion, the square root of AVE was greater than the correlation coefficients among all variables, establishing discriminant validity (see Table 3). In addition, the HTMT (Heterotrait-Monotrait Ratio) value was in the range of 0.373–0.811 ($p < 0.01$), which met the criterion (<0.90) (Henseler et al., 2015).

While only two items were used for communication and renewal intention during the analysis process, prior research supports this approach, showing that two-item reflective scales can be valid when constructs are concrete and demonstrate high reliability (Diamantopoulos et al., 2012; Pedeliento et al., 2016). In our model, both constructs showed CR > 0.90 and AVEs > 0.80.

5.3. Assessment of common method bias

Since the data were collected from the same respondents, common method bias (CMB) was evaluated using both procedural and statistical approaches. First, the procedural approach involved: (a) identifying and correcting wording errors through a pre-test, (b) providing clear instructions to respondents on how to complete the questionnaire, and (c) presenting questionnaire items in a non-sequential order, rather than grouping them by independent, mediating, and dependent variables, to prevent respondents from inferring relationships between variables (Lee et al., 2023; Podsakoff et al., 2003). The statistical approach using Harman's single-factor test, marker variable approach, and VIF values utilized the variance inflation factor (VIF) to assess multicollinearity. First, Harman's single-factor test showed that a single factor explained only 37.8 % of the total variance. Second, a marker variable analysis using self-esteem (maximum correlation with key constructs = 0.09) yielded non-significant relationships. Lastly, the VIF values of all variables were 1.666–4.14, which met the criterion (<5.0), indicating that there was no significant problem with CMB (Chen & Qasim, 2021; Goktas & Dirsehan, 2025; Hair Jr et al., 2021).

5.4. Assessment of the model

PLS-SEM is an analytical technique designed to maximize the explanatory power of endogenous variables while minimizing structural errors (Hulland, 1999). We evaluated the research model using the values of R^2, Q^2, VIF, and SRMR. First, the VIF values for all variables ranged from 1.666 to 4.14, which is below the threshold of 5.0, indicating that multicollinearity is not a concern (Hair et al., 2017). Next, the explanatory power (R^2) of trust (0.758), customer satisfaction (0.771), and renewal intention (0.820) exceeds the strong threshold of 0.67, demonstrating high explanatory power (Chin, 1998). The Stone–Gesser (Q^2) values for predictive relevance were 0.748 for trust, 0.683 for satisfaction, and 0.734 for intention to renew, all of which are greater than zero, confirming the model's predictive relevance (see Table 3). Finally, the standardized root mean square residual (SRMR) value was 0.046, which is below the 0.08 benchmark, indicating that the research model has an adequate fit and strong predictive power (Henseler et al., 2015).

5.5. Hypotheses testing

Fig. 3 presents the structural model estimated using PLS-SEM. All

Table 2
Measurement model.

Constructs and items	Factor loadings	α	CR	AVE
Customer relationship-building Behavior - Approach		0.961	0.972	0.895
My insurance consultant contacts me in advance before visits.	0.931			
My insurance consultant adheres to scheduled appointments and time commitments.	0.957			
My insurance consultant behaves politely during visits.	0.956			
My insurance consultant creates a welcoming atmosphere during visits.	0.941			
Customer relationship-building Behavior - Inducement		0.955	0.968	0.882
My insurance consultant provides gifts on occasions such as birthdays, anniversaries, or holidays.	0.929			
My insurance consultant acknowledges and celebrates my personal achievements, such as promotions.	0.937			
My insurance consultant sends congratulatory messages or gifts for significant events.	0.952			
My insurance consultant expresses sympathy during difficult times in my life.	0.938			
Customer relationship-building Behavior - Common Grounding		0.943	0.963	0.897
My insurance consultant and I discuss personal and family-related matters.	0.966			
My insurance consultant and I share information about work and daily life.	0.969			
My insurance consultant and I talk about hobbies, favorite foods, or entertainment preferences.	0.905			
Customer relationship-building Behavior - Networking		0.971	0.976	0.851
My insurance consultant actively joins groups or communities that benefit their business.	0.901			
My insurance consultant organizes and leads gatherings to build personal networks.	0.931			
My insurance consultant connects clients by facilitating meetings and interactions.	0.945			
My insurance consultant creates professional networks to gain valuable information for clients.	0.910			
My insurance consultant uses social media (e.g., KakaoTalk, Instagram) to arrange client meeting.	0.937			
My insurance consultant uses social media to enhance the engagement of client gatherings.	0.922			
My insurance consultant schedules regular meetings through social media.	0.913			
Customer relationship-building Behavior - Communication		0.954	0.977	0.956
My insurance agent frequently provides new and useful information. #				
My insurance agent recommends products that are tailored to my (or my company's) needs. #				
My insurance agent provides timely and accurate policy updates.	0.977			
My insurance agent clearly explains complex insurance terms.	0.978			

(continued on next page)

Table 2 (*continued*)

Constructs and items	Factor loadings	α	CR	AVE
My insurance agent objectively compares various insurance options when introducing products or services. #				
My insurance agent uses up-to-date market information when responding to my questions. #				
Trust		0.977	0.985	0.956
I trust my insurance agent's expertise and service quality.	0.977			
I have confidence in my insurance agent's consultation and contract management.	0.975			
Customers not managed by my insurance agent would also trust their services.	0.981			
Satisfaction		0.98	0.987	0.961
I have a very positive sentiment toward my insurance agent.	0.975			
I have a favorable opinion of my insurance agent's marketing efforts.	0.984			
Overall, I am satisfied with my insurance agent's marketing activities.	0.982			
Renewal intention		0.984	0.992	0.985
If I had the chance, I would be willing to purchase insurance again from my agent. #				
If I had the chance, I would purchase insurance again from my agent. #				
If I had the chance, I would likely join my insurance agent.	0.992			
If I had the chance, I would likely join my insurance agent again.	0.992			

CR: Composite reliabilities, AVE: Average variance extracted.
The items were deleted during the measurement model analysis.

hypothesized paths, standardized coefficients, and significance levels were derived from the bootstrapping analysis.

As shown in Fig. 3 and Table 4, the findings show that approach ($\beta = 0.137$, $t = 2.709$, $p < 0.01$), common grounding ($\beta = 0.240$, $t = 3.916$, $p < 0.01$), and communication ($\beta = 0.528$, $t = 8.270$, $p < 0.01$) positively influence trust, supporting H1a, H1c, and H1e. Meanwhile, inducement ($\beta = 0.067$, $t = 1.420$, n.s.) and networking ($\beta = 0.037$, $t = 0.908$, n.s.) did not influence trust, not supporting H1-b and H1-d.

The findings show that while approach, common grounding, and networking did not influence customer satisfaction (not supporting H2a, H2c, and H2d), inducement ($\beta = 0.089$, $t = 1.671$, $p < 0.10$) had a marginally significant positive effect, thereby supporting H2b. Meanwhile, communication ($\beta = 0.261$, $t = 3.829$, $p < 0.01$) positively influenced customer satisfaction, supporting H2e.

The findings reveal that trust positively influences satisfaction ($\beta = 0.545$, $t = 7.805$, $p < 0.001$) and renewal intention ($\beta = 0.523$, $t = 6.291$, $p < 0.001$), supporting H3 and H4. Lastly, as expected, satisfaction positively influences renewal intention ($\beta = 0.523$, $t = 6.316$, $p < 0.001$), supporting H5.

5.6. Mediation test

We tested the mediating effects of trust and satisfaction in the relationship between relational-building behaviors and renewal intention. Table 5 shows that emotional attachment play a full mediator in the relationship between approach and satisfaction, and common grounding and satisfaction, because the direct effect of approach and common grounding on satisfaction is not significant, the indirect effects of approach ($\beta = 0.074$, $t = 2.631$, $p < 0.01$) and common grounding ($\beta = 0.131$, $t = 3.606$, $p < 0.01$) on satisfaction through trust are significant, respectively and there is no '0' between LLCI and ULCI. On the other hand, trust plays a partial mediator in the relationship between communication and satisfaction, because the direct effect of communication on satisfaction is significant ($\beta = 0.249$, $t = 3.801$, $p < 0.001$), the indirect effect of communication on satisfaction through trust also is significant ($\beta = 0.288$, $t = 5.249$, $p < 0.001$) and there is no '0' between LLCI and ULCI. Lastly, satisfaction plays a partial mediator in the relationship between trust and renewal intention, because the direct effect of trust on renewal intention is significant ($\beta = 0.523$, $t = 6.291$, $p < 0.001$), the indirect effect of trust on renewal intention through satisfaction also is significant ($\beta = 0.285$, $t = 5.184$, $p < 0.001$) and there is no '0' between LLCI and ULCI.

5.7. Model comparison

For model comparison, we examined an alternative model in which relational-building behaviors directly affect renewal intention. Following Ringle et al. (2024) guideline, we used BIC (Bayesian information criterion) and Akaike weight in the model comparison of Smartpls 4.1 to check whether the proposed model is better than the alternative model in which relational-building behaviors directly affect renewal intention. In terms of BIC, Table 6 shows that the average BIC value of the proposed model (−357.264) is closer to 0 than that of the alternative model (−360.808), which means that the proposed model explains the model better than the alternative model. And the Akaike weight value of the proposed model (0.338) is lower than that of the alternative model (0.662), which means that the average Akaike weight value is closer to 0, which means that the proposed model explains the model better than the alternative model. In addition, model comparison between the proposed model and the alternative model was performed according to the alternative model verification guidelines of Morgan and Hunt (1994). The proposed model has 7 significant paths (53.8 %) out of 13 paths, but the alternative model has 9 significant paths (50.0 %) out of 18 paths, indicating that the proposed model has a higher significant path ratio than the alternative model, indicating that the proposed model is superior.

Table 3
Fornell-Larcker criterion/Heterotrait-Monotrait Ratio (HTMT).

	Constructs	1	2	3	4	5	6	7	8
1	Approach	**0.946**							
2	Inducement	0.440/0.455	**0.939**						
3	Common Grounding	0.468/0.485	0.768/0.735	**0.947**					
4	Networking	0.224/0.225	0.514/0.427	0.624/0.527	**0.923**				
5	Communication	0.642/0.668	0.651/0.68	0.701/0.808	0.415/0.653	**0.978**			
6	Trust	0.604/0.62	0.680/0.859	0.757/0.701	0.478/0.784	0.829/0.482	**0.978**		
7	Satisfaction	0.585/0.599	0.662/0.829	0.691/0.682	0.479/0.715	0.802/0.483	0.856/0.874	**0.981**	
8	Renewal intention	0.556/0.568	0.707/0.855	0.748/0.726	0.469/0.772	0.828/0.472	0.873/0.89	0.872/0.888	**0.992**
	Mean	5.498	4.619	3.362	3.394	3.12	4.461	4.402	4.433
	SD	1.618	1.795	2.133	2.083	1.884	1.901	1.888	2.082

Bold: the square root of the variance shared between the constructs and their measures (AVE).

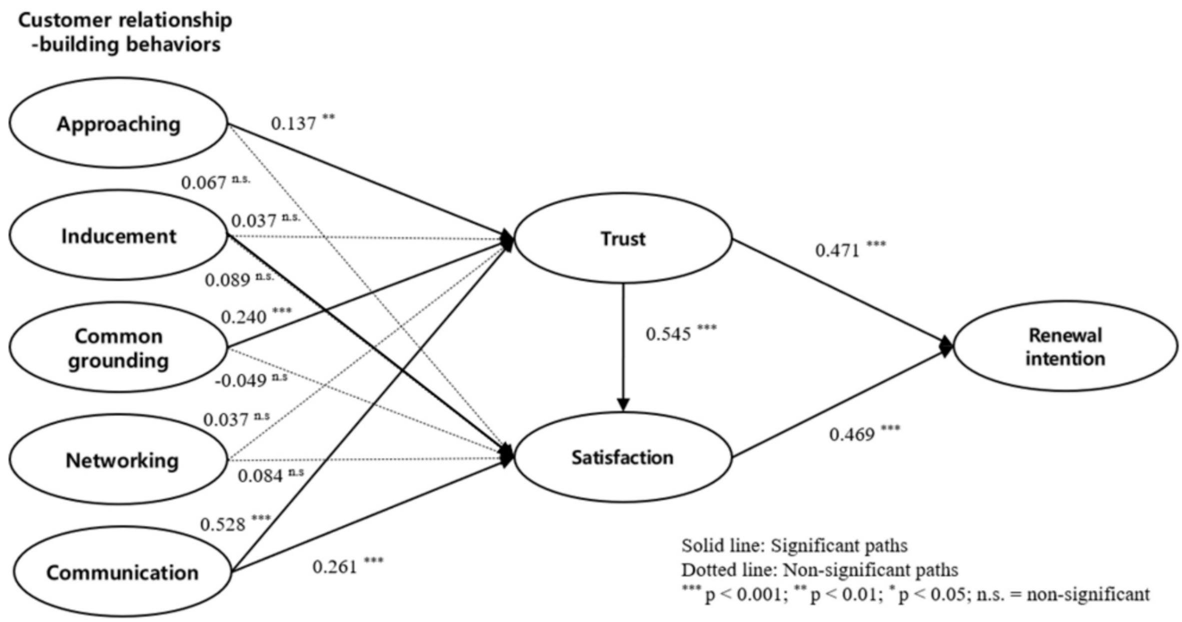

Fig. 3. The estimated structural model (PLS).

Table 4
Structural model.

	Paths	Estimate	t-value	p	Results
H1a	Approaching → Trust	0.137	2.709	0.007 ***	Supported
H1b	Inducement → Trust	0.067	1.420	0.156 n.s.	Not Supported
H1c	Common grounding → Trust	0.240	3.916	0.000 ***	Supported
H1d	Networking → Trust	0.037	0.908	0.364 n.s.	Not Supported
H1e	Communication → Trust	0.528	8.270	0.000 ***	Supported
H2a	Approaching → Satisfaction	0.068	1.197	0.231 n.s.	Not Supported
H2b	Inducement → Satisfaction	0.089	1.920	0.055 ***	Supported
H2c	Common grounding → Satisfaction	−0.049	0.820	0.412 n.s.	Not Supported
H2d	Networking → Satisfaction	0.084	1.632	0.103 n.s.	Not Supported
H2e	Communication → Satisfaction	0.261	3.829	0.000 ***	Supported
H3	Trust → Satisfaction	0.545	7.805	0.000 ***	Supported
H4	Trust → Renewal intention	0.523	6.291	0.000 ***	Supported
H5	Satisfaction → Renewal intention	0.523	6.316	0.000 ***	Supported
		R^2	Q^2		
	Trust	0.758	0.748		
	Satisfaction	0.771	0.683		
	Intention to Renewal Contract	0.820	0.734		
	SRMR	0.046			

* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

5.8. Applying NCA

We examined the impact of five customer relationship-building behaviors of insurance agents on trust, satisfaction, and renewal intentions by utilizing Necessary Condition Analysis (NCA). NCA was applied to

Table 5
Mediation test using bootstrapping.

Paths	Direct effect	Indirect effect			Mediating role
		Estimates	LLCI	ULCI	
Approach → Trust → Satisfaction	0.057 n.s. (1.197)	0.074 *** (2.631)	0.024	0.137	Full
Inducement → Trust → Satisfaction	0.102 n.s. (1.930)	0.037 n.s. (1.393)	−0.013	0.091	–
Common grounding → Trust → Satisfaction	−0.055 n.s (0.822)	0.131 *** (3.606)	0.069	0.213	Full
Networking → Trust → Satisfaction	0.083 n.s. (1.629)	0.020 n.s (0.910)	−0.023	0.065	–
Communication → Trust → Satisfaction	0.249 *** (3.801)	0.288 *** (5.249)	0.193	0.409	Partial
Trust → Satisfaction → Renewal intention	0.523 *** (6.291)	0.285 *** (5.184)	0.189	0.406	Partial

n.s. = non-significant; LLCI: Lower limit confidence interval, ULCI: upper limit confidence interval,
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

Table 6
Model comparison using BIC and Akaike weight.

	Proposed model		Alternative model	
	BIC	Akaike weight	BIC	Akaike weight
Trust	−324.441	0.512	−324.344	0.488
Satisfaction	−332.006	0.497	−332.034	0.503
Renewal intention	−415.345	0.005	−426.046	0.995
Mean	−357.264	0.338	−360.808	0.662

complement the sufficiency-based logic of PLS-SEM by uncovering whether specific behaviors are indispensable for achieving the outcomes. While PLS-SEM identifies factors that significantly influence results, NCA determines whether a minimum level of certain behaviors is required to make the outcome possible—thereby offering a more robust and comprehensive analytical perspective (Dul, 2016).

NCA is used to identify necessary and sufficient factors that affect outcome variables by supplementing the existing PLS-SEM analysis (Richter et al., 2020). To avoid introducing additional linear assumptions between predictor and outcome variables, we applied the recommended Ceiling Regression-Free Disposal Hull (CR-FDH) method. This approach is widely used for continuous data or scenarios where a linear ceiling assumption is appropriate. Next, we assess whether the necessary condition is large enough to be taken seriously by using an effect size measure (d). The NCA effect sizes are evaluated as follows: $0 < d < 0.1$ = small effect, $0.1 \leq d < 0.3$ = medium effect, $0.3 \leq d < 0.5$ = large effect, and $d \geq 0.5$ = very large effect (Dul, 2016).

Table 7 indicates that approach and common grounding values are significant determinants, but not a necessary condition on trust. This indicates that upward changes in the independent variables of approach and common grounding increase trust, but no minimum level of these independent variables is required for the effect on trust to become apparent. Meanwhile, communication is a significant determinant and a necessary condition on trust. This means that an increase in communication increases trust, but a minimum level of communication must be met to achieve a specific value of trust, as indicated in the NCA (Necessary Condition Analysis) bottleneck tables (Table 7). On the other hand, inducement and networking are not significant determinants and necessary conditions of trust. This indicates that a specific threshold level of the exogenous construct, as detailed in the NCA bottleneck tables (Table 8), is essential for the outcome to manifest. Once this threshold is reached, however, further increases in the networking are not advised, as it will not yield additional improvements in the trust improvement.

Table 7 shows that communication and trust are significant determinants and necessary conditions on renewal intention, because communication (estimate = 0.261, $p < 0.001$; $d = 0.177$) and trust (estimate = 0.545, $p < 0.001$; $d = 0.249$) directly influence renewal intention, and their effect sizes are >0.1. In addition, Table 7 shows that communication, trust, and satisfaction are significant determinants and necessary conditions on renewal intention, because communication (estimate = 0.564, $p < 0.001$; $d = 0.229$), trust (estimate = 0.471, $p < 0.001$; $d = 0.105$), and satisfaction (estimate = 0.469, $p < 0.001$; $d = 0.111$) directly influences renewal intention, and their effect sizes are >0.1. These findings demonstrate that certain relationship-building behaviors, particularly communication and trust, are not only statistically significant but also psychologically non-negotiable—offering meaningful insights for practice and theory.

This means that an increase in communication, trust, and satisfaction increases renewal intention, but a minimum level of communication, trust, and satisfaction must be met to achieve a specific value of renewal intention, as indicated in the NCA (Necessary Condition Analysis) bottleneck tables (Table 8).

Lastly, we performed a bottleneck analysis to identify the minimum level of independent variables that is required to obtain a certain level of trust, satisfaction, and renewal intention, respectively (see Table 8). According to Table 7, achieving 80 % renewal intention requires the following minimum levels: approach (0.000 %), communication (20.319 %), inducement (0.000 %), common grounding (0.000 %), networking (0.000 %), trust (13.147 %), and satisfaction (15.936 %). However, to attain a maximum renewal intention level of 100 %, certain minimum thresholds must be met, including approach (0.000 %), communication (35.458 %), inducement (0.000 %), common grounding (20.717 %), networking (0.000 %), trust (36.653 %), and satisfaction (24.303 %). Achieving a high level of renewal intention will not be feasible without first meeting these minimum levels.

While PLS-SEM identifies statistically significant predictors that are sufficient to influence the outcome, NCA determines whether specific conditions are fundamentally necessary for the outcome to materialize. By integrating these two approaches, our study examines both the strength and indispensability of relationship-building behaviors. For instance, communication and trust were not only confirmed as statistically significant through PLS-SEM but also recognized as essential conditions via NCA. This dual-method framework provides deeper theoretical insights and more actionable practical guidance.

6. Discussion

This study explores the impact of insurance agents' relationship-building behaviors on trust, satisfaction, and renewal intention in high-information-asymmetry services. Using PLS-SEM and NCA, we identified both significant and necessary predictors of customer loyalty. Traditional PLS-SEM is concerned primarily with sufficiency logic (i.e., if X is present, then Y is likely to occur). In contrast, NCA supplements this approach by examining necessity logic (i.e., X must be present for Y to occur) (Dul, 2016; Richter et al., 2020). By integrating both methods, researchers can achieve a more thorough and comprehensive understanding of causal relationships.

The findings reveal that communication, inducement, common-grounding, networking, and information sharing play a crucial role in shaping trust and satisfaction, which, in turn, drive renewal intention. Notably, communication emerged as a necessary condition, highlighting its fundamental role in customer relationship development. Our results align with prior research on the established link between relationship quality and loyalty (Rather & Hollebeek, 2019; Chiu et al., 2021), while extending these insights by validating the minimum behavioral thresholds required for customer retention.

The integrated use of PLS-SEM and NCA provides a multidimensional analytical framework, reinforcing both the robustness and practical relevance of our findings. Moreover, behavioral patterns—such as

Table 7
PLS-SEM and NCA (CR-FDH) effect sizes (*d*).

Constructs	Trust		Satisfaction		Renewal intention	
	PLS-SEM	CR-FDH	PLS-SEM	CR-FDH	PLS-SEM	CR-FDH
Approach	0.137 **	0.033	0.068 n.s	0.000	**0.146** **	0.000
Communication	0.528 ***	0.101	0.261 ***	0.177	**0.564** ***	0.229
Inducement	0.087 n.s.	0.000	0.089 n.s.	0.000	**0.101** *	0.000
Common grounding	0.240 ***	0.012	−0.049 n.s.	0.012	**0.168** **	0.012
Networking	0.037 n.s	0.000	0.084 n.s	0.000	**0.074** n.s.	0.000
Trust			0.545 ***	0.249	**0.471** ***	0.105
Satisfaction					0.469***	0.111

n.s. = non-significant; Bold: The values of total effects using PLS-SEM bootstrapping.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

Table 8
Bottle neck table (CF-FDH).

%	Tru					Sat						Reint						
	Appr	Comm	Indu	Cog	Net	Appr	Comm	Indu	Cog	Net	Tru	Appr	Comm	Indu	Cog	Net	Tru	Sat
0	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN
10	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN
20	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN
30	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN
40	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	NN	5.976	NN	NN	NN	NN	NN
50	NN	4.382	NN	NN	NN	NN	4.382	NN	NN	NN	7.171	NN	9.96	NN	NN	NN	NN	5.179
60	NN	5.976	NN	NN	NN	NN	5.976	NN	NN	NN	13.147	NN	12.351	NN	NN	NN	NN	5.179
70	NN	5.976	NN	NN	NN	NN	12.351	NN	NN	NN	21.912	NN	17.131	NN	NN	NN	6.773	13.944
80	NN	9.96	NN	NN	NN	NN	20.319	NN	NN	NN	34.661	NN	20.319	NN	NN	NN	13.147	15.936
90	3.187	12.351	NN	17.53	NN	NN	30.279	NN	17.53	NN	43.825	NN	30.279	NN	17.53	NN	21.912	22.311
100	8.367	17.131	NN	20.717	NN	NN	45.02	NN	20.717	NN	75.697	NN	35.458	NN	20.717	NN	36.653	24.303

Appr: Approach, Comm: Communication, Indu: Inducement, Cog: Common grounding, Net: Network, Tru: Trust, Sat: Satisfaction, Reint: Renewal intention.

customers' preferred interaction modes and policy renewal behaviors—offer critical contextual insights into how insurance agents' relationship-building efforts translate into customer loyalty. These insights serve as a comprehensive foundation for understanding the mechanisms that drive loyalty and contribute to the advancement of theory and practice in service marketing.

6.1. Theoretical implications

This study examined the impact of insurance agents' relationship-building behaviors on trust, satisfaction, and renewal intention, highlighting the critical role of these factors through necessary condition analysis (NCA).

The key theoretical implications are as follows. First, this research identified that an insurance agent's ability to build relationships with customers is essential for fostering trust and satisfaction. While previous research has primarily examined the effect of trust and satisfaction on renewal decisions, this study underscores the importance of relationship-building as a foundational step in developing trust and satisfaction. From this perspective, the study provides empirical support for the significance of long-term customer relationships as proposed by relationship marketing theory. Additionally, it emphasizes the pivotal role of insurance agents in cultivating trust, reinforcing the principles of trust formation within the frameworks of relationship marketing and social exchange theory.

Second, this study confirmed that communication is the most crucial component of customer relationship-building behavior in fostering trust and satisfaction. The findings from the combined PLS-SEM and NCA analyses indicate that effective communication is a necessary condition for building trust and satisfaction. Customers highly value clear information delivery and ongoing communication in their interactions with insurance agents. These insights contribute to a more detailed understanding of the relationship between the quality of information exchange and trust formation within the framework of relationship marketing research. Third, this study showed that trust and satisfaction are critical factors in determining policy renewal intention. The previously established link between relationship quality and customer loyalty (Morgan & Hunt, 1994) was empirically validated once again, reinforcing the idea that in long-term service relationships such as insurance, trust and satisfaction significantly increase the likelihood of continued policy retention.

Furthermore, NCA analysis revealed that without achieving a certain level of trust and satisfaction, renewal intention does not develop. This finding goes beyond a simple correlation, emphasizing that trust and satisfaction are not just influencing factors but essential conditions for fostering customer loyalty. Finally, this study broadened the application of relationship marketing theory by contextualizing it within the insurance industry. Given the intangible nature of insurance products and their long-term contractual structures, the impact of relationship-building with insurance agents on customer decision-making is particularly significant. By empirically analyzing the effects of relationship-building behaviors in this setting and integrating PLS-SEM and NCA, this study identified the role of essential factors, thereby contributing to the extension of relationship marketing research into a wider range of service industries.

6.2. Managerial implications

First, to enhance trust, it is essential to adopt consistent communication and timely consulting systems that offer customers the information they need in a timely manner. By analyzing policy and claim histories alongside lifestyle data in real time, such systems automate personalized product recommendations and coverage updates, thereby reducing customer waiting times and allowing advisors to concentrate on more specialized consultations. For instance, in the United States, Lemonade leverages a chatbot named Maya and an automated underwriting system called Jim to provide expedited claims processing and high levels of customer satisfaction. In Korea, Samsung Life Insurance employs AI algorithms to assess individual risk profiles and existing policies, thereby implementing a digital consulting service that offers customized insurance solutions. Furthermore, DB Insurance utilizes an AI chatbot known as Customer Bot, providing continuous premium calculations, coverage information, and claims assistance to strengthen contactless counseling. Finally, the inclusion of infographics and videos can further improve customers' comprehension of insurance products and elevate the overall service experience.

Second, insurers can enhance empathic engagement by integrating emotion analytics tools that dynamically adjust consultation styles—ranging from informational to highly empathic—based on real-time vocal cues. By examining variations in tone, pace, and other speech patterns, these systems swiftly identify the most suitable counseling approach. Moreover, incorporating key life events—such as marriage, childbirth, or health checkups—enables AI to recommend personalized insurance solutions aligned with each client's evolving needs. For instance, Kyobo Life Insurance has tested a pilot program combining voice recognition with emotional analysis to detect subtle affective changes during calls, while Ping An (China) computes "emotional scores" to provide customized advisory services. Through these customer insights, insurers can reinforce trust, elevate customer satisfaction, and ultimately bolster long-term renewal intention.

Third, to enhance Approach—that is, the accessibility dimension of customer service—insurers can adopt a hybrid sales model that integrates mobile, video, and face-to-face channels. For example, Hyundai Marine & Fire Insurance introduced an online consulting platform enabling customers to compare and purchase insurance products directly via mobile devices, whereas AXA (France) relies on mobile applications and websites to allow customers to independently explore and select suitable policies. This approach ensures that customers can access consultations at their preferred time and location, while agents can engage with a higher volume of clients within the same time frame, thereby improving operational efficiency.

Fourth, although the study findings indicate a somewhat limited direct effect of networking, segmenting existing customers and leveraging algorithm-based methods to connect prospective clients with insurance agents can significantly amplify the network effect. For instance, insurers might adopt a B2B2C model by partnering with a healthcare company to operate a shared platform that links the healthcare firm's client base to insurance agents. Under such an arrangement, AI would analyze accumulated customer data to automatically match each prospect with the most suitable agent, thus facilitating new customer acquisitions and generating cross-industry collaboration benefits. In practice, Ping An (China) has reportedly developed a B2B2C ecosystem by collaborating with multiple healthcare and financial partners, employing AI algorithms to match customer segments with appropriate agents, which has helped bolster long-term customer retention.

Finally, for effective after-service management, insurers should supply analytics-based insurance consulting that continuously track customers' health status and insurance usage patterns, thereby recommending optimal coverage. For instance, Samsung Life Insurance offers a personalized coverage analysis service enabling immediate coverage updates via mobile or telephone consultations, while AXA has established a system that automatically recommends new products or riders based on changes in a customer's profile (e.g., family composition or job transitions). By delivering rapid feedback, these AI-driven after-service strategies can substantially enhance policy retention and customer loyalty.

In summary, insurance agents should focus on maintaining long-term customer relationships by leveraging digital technology to offer personalized services, improving accessibility through hybrid consulting models, utilizing AI-based customer insights, and strengthening aftercare. This approach not only enhances trust but also improves policy renewal rates and overall loyalty.

6.3. Limitations and future research

The limitations of this study and directions for future research are as follows. First, the study's focus on the insurance industry within a single region limits the generalizability of the findings to other industries and cultural contexts. Future research should conduct comparative analyses across industries and countries to extend the scope of applicability. In addition, examining how contextual factors—such as regional economic conditions, urban-rural differences, and product-specific customer expectations—moderate the relationship between relationship-building behaviors and renewal intention would provide deeper insights. Second, the cross-sectional research design restricted the ability to examine long-term changes in customer relationship formation. A longitudinal study is needed to better understand how trust and satisfaction evolve over time based on service experience. Third, while this study did not find significant differences through MGA (multi-group analysis) when considering gender, future research should explore the moderating effects of other factors such as age, income level, and prior insurance subscription experience. These variables may provide deeper insights into customer relationship dynamics. Fourth, we tried various methods to increase the reliability and validity of constructs, but our data has some composite reliability and Cronbach's α values are very high, which indicates a multicollinearity issue. Therefore, we need to solve this problem in future research by more carefully solving the multicollinearity issue among constructs. Lastly, this study did not assess the role of digital technology in shaping customer relationship-building behaviors. Future research should analyze how emerging technologies—such as AI, big data analytics, and customer experience management systems—influence the formation of trust and satisfaction in the insurance industry.

CRediT authorship contribution statement

Joon-Ho Moon: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Soon-Tae Lee:** Writing – review & editing, Writing – original draft, Validation, Formal analysis, Data curation, Conceptualization. **Yong-Ki Lee:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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